

BANARAS HINDU UNIVERSITY
B.Sc. (Hons.) Semester VI Examination, 2013-14

Botany

Paper: BOB-603 — Cytogenetics and Evolutionary Processes

Time: Three hours

Full Marks: 70

Note: Answer five questions in all, selecting at least two questions each from Section-A and Section-B. Section-C is compulsory. Question No. 9 carries 10 marks and the remaining questions carry 15 marks each. The figures in the right-hand margin indicate marks.

SECTION – A

Question 1. Write an account on the ultra structural organization of chromatin. [15]

Question 2. Define translocation. Describe the cytogenetics of a translocation heterozygote with the help of suitable diagrams. [15]

Question 3. Define aneuploidy. How will you distinguish between: [15]

- (a) double monosomics and nullisomics
- (b) primary trisomies and secondary trisomics

Question 4. What are transitions and transversions? Name few physical mutagens and explain their mode of action. [15]

SECTION – B

Question 5. Write short notes on any two of the following: [7.5 × 2 = 15]

- (a) Base substitution
- (b) Kappa Particle
- (c) Multiple allelism

Question 6. Differentiate between any three pairs of the following: [5 × 3 = 15]

- (a) Allopolyploid and segmental allopolyploidy
- (b) Deletion and duplication
- (c) Crossing over and linkage
- (d) Centromere and Kinetochore

Question 7. Write explanatory note on any two of the following: [7.5 × 2 = 15]

- (a) Heterosis and its evolutionary significance

- (b) Quantitative inheritance
- (c) Trisomies in *Datura stramonium*

Question 8. Write short notes on any two of the following: [7.5 × 2 = 15]

- (a) Organellar genetics
- (b) Components of Karyotype
- (c) Linkage analysis

SECTION – C

Question 9. Answer the following in not more than 50 words each: [1 × 10 = 10]

- (a) Define telomere
- (b) What is constitutive heterochromatin?
- (c) Define Karyotyping
- (d) What is *Raphanobrassica*?
- (e) Role of Colchicine in plants
- (f) What is *in-situ* hybridization?
- (g) Define self sterility in plants?
- (h) What is Runner's effect?
- (i) What do you mean by genetic load?
- (j) Define somatic cell hybrids?

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